

Types of RNN

Contents

1. Overview	2
2. Prerequisites	2
3. Many-to-one	2
4. One-to-many	3
5. Many-to-many	4
5.1 Same length	5
5.2 Different lengths	6
6. One-to-one	7
7. The types in Keras	8
8. Summary	9
9. Sources	9
10. Key terms	9

1. Overview

Key point: RNNs are grouped by two questions: is the input a sequence, and is the output a sequence? The answers give many-to-one, one-to-many and many-to-many (same or different lengths). One-to-one, with no sequence on either side, is an ordinary network, not an RNN.

The [RNN forward propagation Note](#) built one RNN: it reads a review word by word and gives one prediction at the end. The same recurrent layer can be wired in other ways, depending on what goes in and what comes out.

Type	Input	Output	Example
Many-to-one	sequence	one value	sentiment analysis, rating prediction
One-to-many	one value (not a sequence)	sequence	image captioning, music generation
Many-to-many, same length	sequence	sequence, one output per input	part-of-speech tagging, named entity recognition
Many-to-many, different lengths	sequence	sequence of another length	machine translation
One-to-one	one value	one value	image classification (not an RNN)

Each type has its own unrolled picture, shown in Figures 1 to 5. The type also changes how backpropagation runs through the network, which the [backpropagation through time Note](#) covers.

2. Prerequisites

- The [why RNNs Note](#): sequential and non-sequential data.
- The [RNN forward propagation Note](#): the recurrent layer, the hidden state h_t and unfolding through time.
- The [RNN sentiment analysis Note](#): a many-to-one RNN trained in Keras.

3. Many-to-one

Key point: A sequence goes in, one value comes out. The RNN reads every time step and predicts only after the last one.

A **many-to-one** RNN takes a sequence as input, such as a sentence, a string of characters or a time series, and gives a non-sequential output: one number, such as a class label 1 or 0.

- **Sentiment analysis:** the input is a review, read word by word; the output is its sentiment, 1 (positive) or 0 (negative).

- **Rating prediction:** the input is a review; the output is a star rating from 1 to 5. The output is still one value, so this is a multi-class classification problem on top of a many-to-one RNN.

many-to-one: sentiment analysis

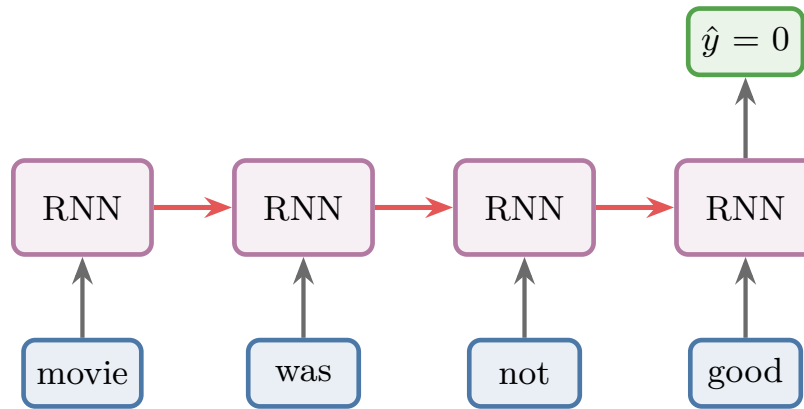


Figure 1: Many-to-one, unrolled: one word enters at each time step, the hidden state passes to the next step (red arrows), and only the last step produces an output

Figure 1 is the RNN of the [RNN forward propagation Note](#), unrolled: many inputs, one output, hence the name. Goodfellow §10.2 describes this pattern as a network that reads an entire sequence and then produces a single output, a fixed-size summary of the sequence.

Python: In Keras a recurrent layer returns only its last hidden state by default (`return_sequences=False`), which is exactly many-to-one.

```
model = keras.Sequential([
    keras.Input(shape=(4, 5)),           # (time steps, features)
    keras.layers.SimpleRNN(3),           # last hidden state only
    keras.layers.Dense(1, activation="sigmoid")])
# input (3, 4, 5) -> output (3, 1): one number per review
```

4. One-to-many

Key point: One non-sequential input goes in, a sequence comes out. The input enters once, and the RNN produces an output at every time step.

A **one-to-many** RNN takes a non-sequential input and produces a sequence. An image is non-sequential data: it is a grid of numbers with no order in time. The output can be words, a sentence or a time series.

- **Image captioning:** the input is one image; the output is a sentence that describes it. An image of a man playing cricket might give “a man plays cricket”.
- **Music generation:** one input starts the network, and it keeps producing music notes, one per time step.

one-to-many: image captioning

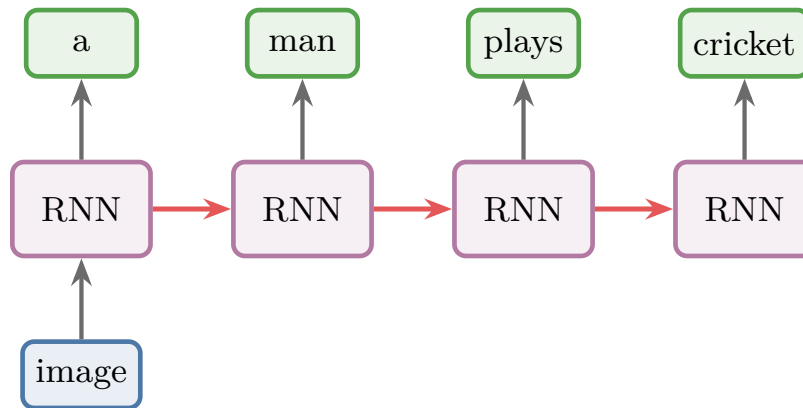


Figure 2: One-to-many, unrolled: the image enters at the first time step only; every time step produces one word of the caption

In Figure 1 the arrows fanned in to one output; in Figure 2 they fan out. The input enters once, at the first time step. At every time step the RNN produces an output and passes its hidden state on to the next step. Goodfellow §10.2.4 names image captioning as the standard task for an RNN that maps one fixed-length vector to a sequence.

Extra: Goodfellow §10.4 gives two ways to feed the single input to such an RNN: as the initial hidden state, or as an input at every time step (and the two can be combined). The Notebook uses the second way: Keras’ `RepeatVector(6)` copies the input to 6 time steps, and the RNN with `return_sequences=True` returns 6 hidden states. An input of shape $(2, 8)$, two “images” of 8 numbers, gives an output of shape $(2, 6, 5)$: 6 words per image, each a probability over 5 words.

5. Many-to-many

Key point: A sequence goes in and a sequence comes out. Such models are also called sequence-to-sequence models. The two sequences can have the same length or different lengths.

A **many-to-many** RNN takes a sequence and returns a sequence, so it is also called a **sequence-to-sequence** (seq2seq) model. It comes in two kinds.

5.1 Same length

Key point: One output for every input: the RNN predicts at every time step.

In a **same-length many-to-many** RNN, the output sequence has as many elements as the input sequence.

- **Part-of-speech tagging:** for each word of a sentence, predict its part of speech. “my name is Riya” gives pronoun, noun, verb, proper noun: 4 words in, 4 tags out. In natural language processing (NLP), tagging is a common preprocessing step.
- **Named entity recognition (NER):** for each word, decide whether it is an entity, a specific thing the program must act on. In “let’s meet at 7 pm at the airport”, “7 pm” and “airport” are entities. Chatbots use NER to pick out times and places.

many-to-many, same length: part-of-speech tagging

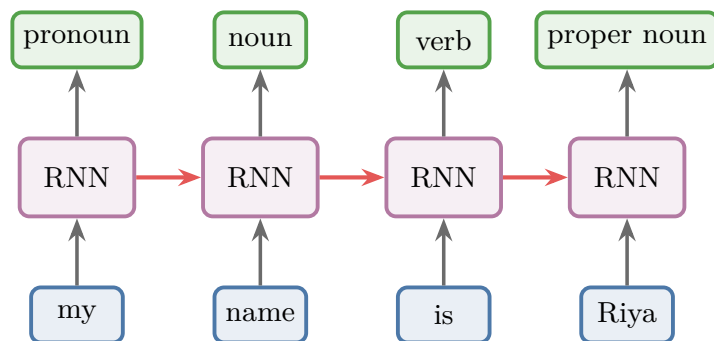


Figure 3: Many-to-many with the same length, unrolled: each time step takes one word and gives one tag

In Figure 3, each time step takes the next input and gives an output, and the hidden state moves on, until the inputs run out. Goodfellow §10.2 lists this as the first RNN pattern: an output at each time step, with recurrent connections between hidden units.

Python: `return_sequences=True` makes the recurrent layer return the hidden state of every time step. A `Dense` layer on that 3D output acts on the last axis, so it makes one prediction per time step (Keras documentation, `Dense`).

```
model = keras.Sequential([
    keras.Input(shape=(4, 5)),
    keras.layers.SimpleRNN(3, return_sequences=True),
    keras.layers.Dense(4, activation="softmax")]) # 4 tags
# input (3, 4, 5) -> output (3, 4, 4): 4 tags per word
```

With the same weights, the last of the hidden states returned by `return_sequences=True` equals the single hidden state returned by `return_sequences=False` (Notebook). The two

settings run the same computation and differ only in what they hand on.

5.2 Different lengths

Key point: The output sequence can be longer or shorter than the input. The RNN first reads the whole input (the encoder), and only then starts writing the output (the decoder).

In a **variable-length many-to-many** RNN, the input and output sequences can have different lengths. The standard example is **machine translation**: translating a sentence from one language into another. Nothing guarantees that a sentence and its translation have the same number of words: some languages say the same thing in fewer words. “I am going home” has 4 words; its Hindi translation “main ghar ja raha hoon” has 5.

many-to-many, different lengths: machine translation (English to Hindi)

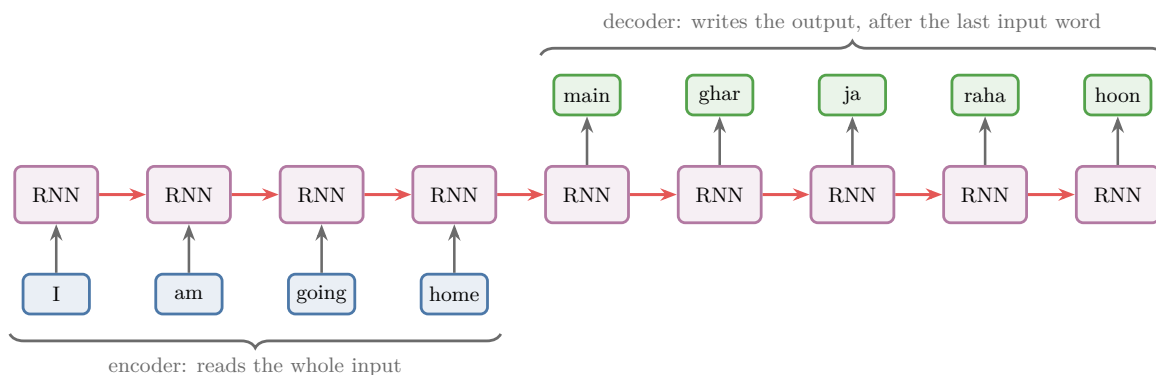


Figure 4: Many-to-many with different lengths, unrolled: the encoder reads all 4 English words; only after the last one does the decoder produce the 5 Hindi words

In Figure 4 the network produces nothing while it reads the input. It starts producing output only after the last input word. The reason is the task: translation does not work word by word. We read the whole sentence, understand its meaning and grammar, and only then decide on the translation. Translating each word as soon as it arrives would lose both the grammar and the context.

The two parts have names.

- The **encoder** is the part that reads the input. It emits a summary of the input, usually a simple function of its last hidden state (Goodfellow §10.4).
- The **decoder** is the part that writes the output, starting from that summary.

Together they form an **encoder-decoder** architecture. Its advance over the earlier patterns is that the input length and the output length can differ (Goodfellow §10.4). Speech recognition and question answering have the same need (Goodfellow §10.4).

Extra: A minimal encoder-decoder in Keras (Notebook): a `SimpleRNN(3)` encoder keeps only its last hidden state; `RepeatVector(6)` hands it to 6 time steps; a `SimpleRNN(3, return_sequences=True)` decoder and a `Dense(7, activation="softmax")` layer produce 6 words from a 7-word vocabulary. An input of shape $(3, 4, 5)$ gives an output of shape $(3, 6, 7)$: 4 input words become 6 output words. The model is untrained; it only shows the shapes.

6. One-to-one

Key point: Non-sequential in, non-sequential out: no time steps and no recurrence. One-to-one is an ordinary ANN or CNN, so technically it is not an RNN.

A **one-to-one** network takes a non-sequential input and gives a non-sequential output. Image classification is an example: the input is an image, a grid of values, and the output is 1 or 0, cat or not.

one-to-one: image classification

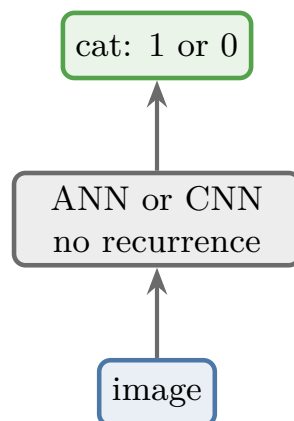


Figure 5: One-to-one: one input, one output, no time steps and no feedback

In Figure 5 everything happens in a single time step, with no feedback connection. Such a network is the plain neural network of the earlier Notes, an ANN or a CNN. One-to-one is listed only to complete the picture; technically there are three types of RNN: many-to-one, one-to-many and many-to-many, the last with its two kinds.

7. The types in Keras

Key point: Two switches build every type: `return_sequences` decides whether the recurrent layer hands on one hidden state or all of them, and `RepeatVector` turns one vector into a sequence.

The Notebook builds one small model of each type, untrained, and checks its shapes. The input is the batch of three one-hot reviews from the [RNN forward propagation Note](#), shape (3, 4, 5), or two “images” of 8 numbers, shape (2, 8).

Type	Layers	Input shape	Output shape
Many-to-one	<code>SimpleRNN(3), Dense(1)</code>	(3, 4, 5)	(3, 1)
Many-to-many, same length	<code>SimpleRNN(3, return_sequences=True), Dense(4)</code>	(3, 4, 5)	(3, 4, 4)
One-to-many	<code>RepeatVector(6), SimpleRNN(3, return_sequences=True), Dense(5)</code>	(2, 8)	(2, 6, 5)
Many-to-many, different lengths	<code>SimpleRNN(3), RepeatVector(6), SimpleRNN(3, return_sequences=True), Dense(7)</code>	(3, 4, 5)	(3, 6, 7)
One-to-one	<code>Dense(3), Dense(1)</code>	(2, 8)	(2, 1)

A 3D output (batch, time steps, values) means one prediction per time step; a 2D output (batch, values) means one prediction per observation.

8. Summary

Type	Input	Output	Unrolled shape	Example
Many-to-one	sequence	one value	output at the last step only	sentiment analysis
One-to-many	one value	sequence	input at the first step, output at every step	image captioning
Many-to-many, same length	sequence	sequence, same length	input and output at every step	part-of-speech tagging, NER
Many-to-many, different lengths	sequence	sequence, other length	encoder reads all, then decoder writes	machine translation
One-to-one	one value	one value	one step, no recurrence	image classification

- RNN types are named by whether the input and the output are sequences.
- Many-to-one reads the whole sequence and predicts once; Keras' default `return_sequences=False`.
- One-to-many turns one input into a sequence, as in image captioning.
- Many-to-many (sequence-to-sequence) predicts at every step when the lengths match, and uses an encoder and a decoder when they differ.
- One-to-one has no time steps, so it is an ordinary ANN or CNN, not an RNN.

9. Sources

- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. Chapter 10: §10.2 (design patterns for recurrent networks, single output at the end of the sequence), §10.2.4 (vector-to-sequence RNNs, image captioning), §10.4 (encoder-decoder sequence-to-sequence architectures). deeplearningbook.org/contents/rnn.html.
- Keras API documentation: `SimpleRNN` (`return_sequences`), `RepeatVector` and `Dense` (inputs of rank greater than 2) layers, keras.io/api/layers/.

10. Key terms

Term	Meaning
Many-to-one RNN	An RNN that reads a sequence and gives one output at the end
One-to-many RNN	An RNN that takes one non-sequential input and produces a sequence
Many-to-many RNN	An RNN that takes a sequence and produces a sequence; also called sequence-to-sequence

Term	Meaning
Sequence-to-sequence (seq2seq) model	A model whose input and output are both sequences
Same-length many-to-many	Many-to-many with one output per input time step
Variable-length many-to-many	Many-to-many whose output length can differ from its input length
One-to-one	A network with non-sequential input and output: an ordinary ANN or CNN, not an RNN
Encoder	The part of a sequence-to-sequence model that reads the whole input and summarises it
Decoder	The part of a sequence-to-sequence model that writes the output from the encoder's summary
Part-of-speech tagging	Labelling every word of a sentence with its part of speech
Named entity recognition (NER)	Marking the words of a sentence that are entities, such as times and places
Machine translation	Translating a sentence from one language into another
Image captioning <code>return_sequences</code>	Producing a sentence that describes an image Keras switch: False returns the last hidden state, True returns the hidden state of every time step
<code>RepeatVector</code>	Keras layer that repeats one vector at several time steps